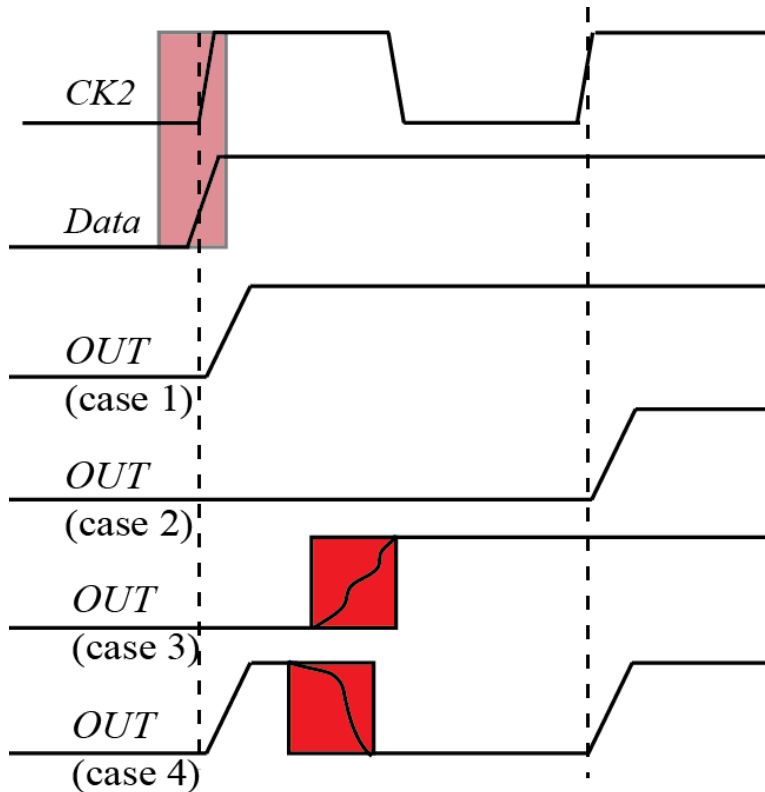


Consequences of Forbidden Event



Case 1

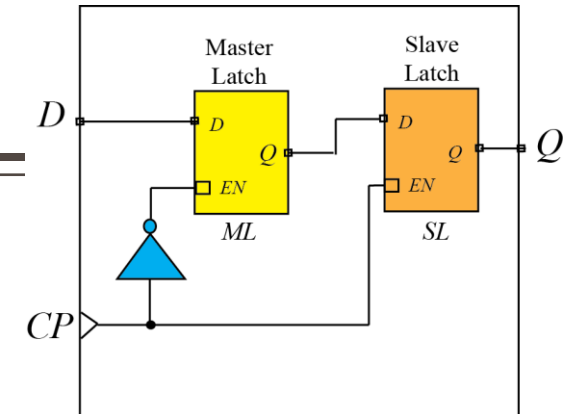
- ML latches the data properly or it enters into a metastable state and quickly resolves to stable 1
- SL latches the data correctly and produces a stable 1

Case 2

- ML misses the data completely or it enters into a metastable state and quickly resolves to 0.
- SL latches the data correctly in the next clock cycle

Case 3

- ML goes into a metastable state. SL produces a 1 (pull-up)
- Then, ML resolves to 0 and SL follows it by toggling its output value.
- In the next cycle, both ML and SL produce the data correctly.



Case 3

- ML goes into a metastable state and resolves to 1 within the first clock cycle
- SL produces a metastable output initially, but it resolves to 1 within the first cycle
- Effective clock to Q delay is significantly high

Summary

- Behavior unpredictable (cycle of data capture and delay)